



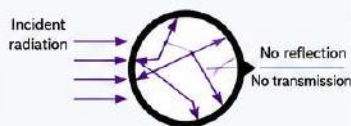
BLACK BODY RADIATION



A black body is an ideal body that absorbs all radiations incident on it and does not reflect or transmit any radiation. When a black body is heated, it emits radiations of all wavelengths. The study of this radiation is called black body radiation.

1. WHAT IS BLACK BODY?

A black body is an ideal physical system which absorbs all the radiation falling on it and does not reflect, transmit or scatter any radiation.



- Has a small hole to allow radiation to enter and leave.
- Absorptivity (α) = 1, Reflectivity (ρ) = 0, Transmissivity (τ) = 0

A black body is a perfect absorber and a perfect emitter.

2. EXPERIMENTAL OBSERVATIONS

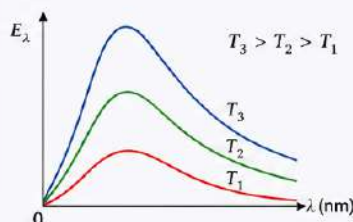
- When a black body is heated, it emits continuous radiation.
- The color of the emitted radiation changes from red (at lower temperature) to white (at higher temperature).
- The emitted radiation depends on the temperature of the black body.



Black body radiation is continuous and covers the entire range of wavelengths (from ultraviolet to infrared).

3. BLACK BODY RADIATION CURVES

The variation of energy emitted (E) with wavelength (λ) at different temperatures is shown below.



- As temperature increases, energy emitted at all wavelengths increases.
- Peak of the curve shifts towards shorter wavelength with increase in temperature.
- Area under the curve increases with temperature.

4. IMPORTANT LAWS

(i) Wien's Displacement Law

The product of wavelength at maximum energy ($\lambda_{\max}$) and absolute temperature (T) is constant.

$$\lambda_{\max} T = b \quad (b = 2.898 \times 10^{-3} \text{ m K})$$

(ii) Stefan-Boltzmann Law

The total energy radiated per unit area per second (E) is proportional to the fourth power of the absolute temperature.

$$E = \sigma T^4 \quad (\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4})$$

(iii) Rayleigh-Jeans Law (Classical)

At longer wavelengths (low frequencies),

$$E \propto \frac{T}{\lambda^4} \quad (\text{Failed at short wavelengths})$$

SIGNIFICANCE

- ✓ Helps in understanding the nature of thermal radiation.
- ✓ Provided evidence for the quantum theory of radiation.
- ✓ Basis for development of Planck's quantum theory.
- ✓ Applications in astrophysics, pyrometry, infrared technology, etc.

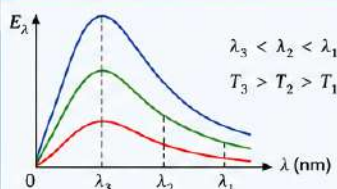
5. STEFAN-BOLTZMANN LAW (DERIVATION IDEA)

If area under the curve at temperature T_1 is E_1 and at T_2 is E_2 ,

$$\text{then } \frac{E_2}{E_1} = \left(\frac{T_2}{T_1}\right)^4 \Rightarrow E \propto T^4$$

Total radiant energy emitted by a black body is proportional to the fourth power of its absolute temperature.

6. WIEN'S DISPLACEMENT LAW (GRAPHICAL)



As temperature increases, $\lambda_{\max}$ decreases (peak shifts left).

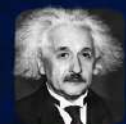
QUICK SUMMARY

Aspect	Description
Nature of radiation	Continuous (all wavelengths)
Depends on	Temperature of black body
Key laws	Wien's Displacement Law, Stefan-Boltzmann Law
Peak shift	Towards shorter wavelength with increase in temperature
Total energy	Proportional to T^4
Significance	Foundation of quantum theory, various applications



REMEMBER

- Black body absorbs all radiation.
- It emits continuous radiation.
- $\lambda_{\max} T = \text{constant}$
- $E_{\text{total}} \propto T^4$



Albert Einstein
(1879–1955)

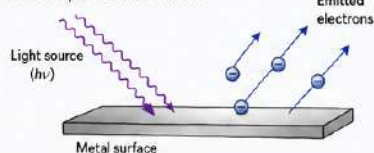
PHOTOELECTRIC EFFECT

Light behaves like a stream of particles (photons) — not just like a wave.

"The photoelectric effect provides compelling evidence for the particle nature of light and was a key step in the development of quantum theory."

1. WHAT IS PHOTOELECTRIC EFFECT?

The emission of electrons from a metal surface when electromagnetic radiation of suitable frequency (above a threshold) falls on it is called the photoelectric effect.



Electrons are emitted only when frequency of incident light is greater than the threshold frequency.

2. OBSERVATIONS (EXPERIMENTAL FACTS)

- For a given metal, electrons are emitted only when frequency (ν) of incident light is greater than a certain minimum value called threshold frequency (ν_0). No time lag is observed.
- The maximum kinetic energy ($K_{\max}$) of photoelectrons depends on the frequency (ν) of incident light and not on its intensity.
- The number of photoelectrons emitted (photoelectric current) is directly proportional to the intensity of incident light, for $\nu > \nu_0$.
- The photoelectric current is almost independent of the potential difference between the electrodes once it is above a certain value (saturation current).

3. EXPLANATION (EINSTEIN'S PHOTOELECTRIC EQUATION)

According to Einstein, light consists of particles called photons, each having energy $h\nu$.

When a photon is incident on a metal surface, it transfers its entire energy to a single electron. Part of this energy is used to overcome the work function (ϕ) of the metal and the rest appears as kinetic energy of the emitted electron.

$$h\nu = \phi + K_{\max}$$

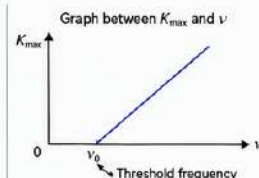
where,

h = Planck's constant (6.626×10^{-34} J s)

ν = frequency of incident light (s^{-1})

ϕ = work function of the metal (J)

$K_{\max}$ = maximum kinetic energy of photoelectron (J)

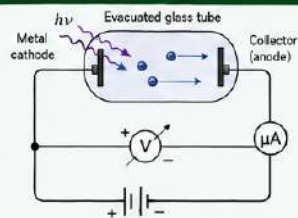


Threshold frequency,

$$\nu_0 = \frac{\phi}{h}$$

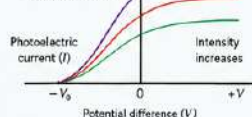
No electrons are emitted for $\nu < \nu_0$.

4. EXPERIMENTAL SETUP AND RESULTS



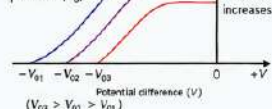
(i) Effect of intensity (at constant $\nu > \nu_0$)

Higher intensity
→ more electrons
→ higher current



(ii) Effect of frequency (at constant intensity)

Higher frequency →
more K.E. →
more stopping potential (V_0)



(iii) Stopping potential (V_0)

The negative potential required to stop the most energetic electrons (photoelectric current becomes zero).

$$eV_0 = K_{\max}$$

where, e = charge on electron
(1.602×10^{-19} C)

5. KEY TERMS

- Threshold frequency (ν_0):** Minimum frequency required to eject electrons.
- Work function (ϕ):** Minimum energy needed to remove an electron from the metal surface.
- Photoelectrons:** Electrons emitted from the metal surface.
- Stopping potential (V_0):** Potential that just stops the fastest photoelectrons.

6. SIGNIFICANCE OF PHOTOELECTRIC EFFECT

Proves Particle Nature of Light



Shows that light can behave as particles (photons), not just as waves.

Foundation of Quantum Theory



Helped in the development of quantum mechanics and modern physics.

Determination of Planck's Constant



Planck's constant (h) can be determined experimentally using photoelectric effect.

Work Function Measurement



Enables the calculation of work function of metals, important in materials science.

Applications in Technology



Used in devices like photocells, solar cells, TV camera tubes, night vision devices, etc.

Basis of Modern Electronics



Principle behind many photoelectric sensors, detectors and electronic instruments.

In Short

The photoelectric effect shows that light consists of discrete packets of energy called photons. Einstein's equation $h\nu = \phi + K_{\max}$ explains all experimental observations successfully and is a milestone in the birth of quantum mechanics.

Remember the Principle:

$$h\nu = \phi + K_{\max}$$

Light is not just a wave...
it is also a particle!



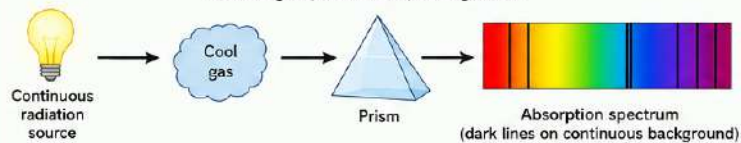
ABSORPTION AND EMISSION SPECTRA



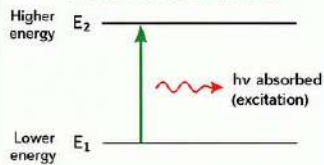
When electrons in atoms or molecules move between energy levels, they absorb or emit radiation of specific wavelengths. These radiations produce **ABSORPTION** and **EMISSION SPECTRA**.

1. ABSORPTION SPECTRA

An absorption spectrum is produced when a substance absorbs radiation from a continuous source. It appears as dark lines on a bright (continuous) background.



ENERGY LEVEL DIAGRAM

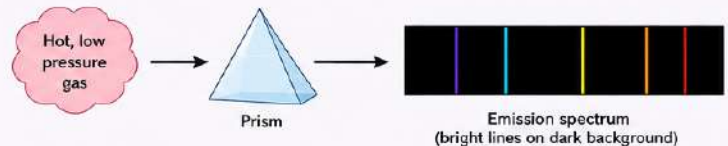


FEATURES

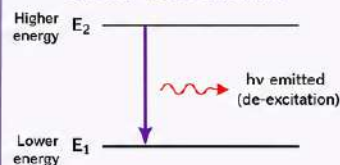
- Dark lines at specific wavelengths.
- Occurs when electrons absorb energy and move to higher energy levels.
- Each element has a unique absorption spectrum.

2. EMISSION SPECTRA

An emission spectrum is produced when a substance in the excited state emits radiation. It appears as bright lines on a dark background.



ENERGY LEVEL DIAGRAM



FEATURES

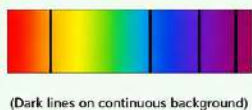
- Bright lines at specific wavelengths.
- Occurs when electrons lose energy and move to lower energy levels.
- Each element has a unique emission spectrum.

SIGNIFICANCE

- ✓ Help in the identification of elements.
- ✓ Used in determining the composition of stars and celestial bodies.
- ✓ Useful in analytical chemistry for qualitative analysis.
- ✓ Provide evidence for quantized energy levels in atoms.

EXAMPLE: HYDROGEN ATOM

ABSORPTION SPECTRUM



(Dark lines on continuous background)

EMISSION SPECTRUM



(Bright lines on dark background)

Hydrogen shows dark lines in its absorption spectrum and bright lines in its emission spectrum at the same wavelengths. These lines correspond to transitions between energy levels.

QUICK SUMMARY

Aspect	Absorption Spectrum	Emission Spectrum
Appearance	Dark lines on bright (continuous) background	Bright lines on dark background
Process	Absorption of radiation	Emission of radiation
Electron transition	Lower $\rightarrow$ Higher energy level	Higher $\rightarrow$ Lower energy level
Background	Bright (continuous)	Dark
Example	Solar spectrum	Hydrogen gas in discharge tube

REMEMBER

Absorption and emission spectra are the fingerprints of elements, revealing their unique energy level transitions.



BOHR'S THEORY OF HYDROGEN ATOM



POSTULATES OF BOHR'S THEORY

1. Stationary Orbits

The electron revolves around the nucleus only in certain fixed orbits called stationary orbits without radiating energy.

2. Quantization of Angular Momentum

Only those orbits are allowed for which the angular momentum of the electron is an integral multiple of $h/2\pi$.

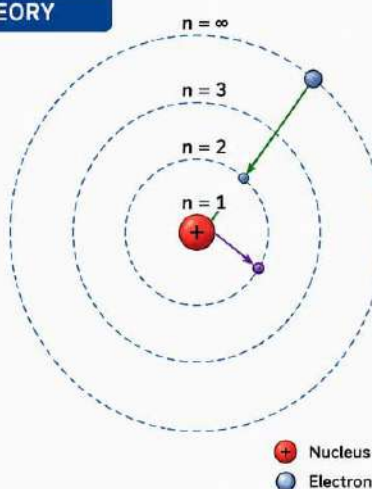
$$mvr_n = \frac{nh}{2\pi} \quad (n = 1, 2, 3, \dots)$$

3. Radiation is Emitted or Absorbed

Radiation is emitted or absorbed only when an electron jumps from one stationary orbit to another.

$$h\nu = |E_i - E_f|$$

where h = Planck's constant, ν = frequency



TRANSITIONS AND SPECTRAL LINES

Energy Levels

$$n = \infty \text{ --- } E = 0$$

$$n = 4 \text{ --- } E_4 = -0.85 \text{ eV}$$

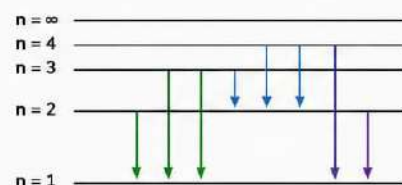
$$n = 3 \text{ --- } E_3 = -1.51 \text{ eV}$$

$$n = 2 \text{ --- } E_2 = -3.40 \text{ eV}$$

$$n = 1 \text{ --- } E_1 = -13.6 \text{ eV}$$

Energy ↑

When electron jumps from higher energy level to lower energy level, a photon is emitted.



Spectral Series (Examples)

Lyman Series : Transitions ending at $n = 1$ (UV region)

Balmer Series : Transitions ending at $n = 2$ (Visible region)

Paschen Series : Transitions ending at $n = 3$ (IR region)

ACHIEVEMENTS



Explained Stability of Atom

Explained why the electron in hydrogen atom does not radiate energy while revolving around the nucleus.



Explained Line Spectrum of Hydrogen

Successfully explained the emission spectrum of hydrogen atom (Lyman, Balmer, Paschen series) using energy levels.

$$E_n = \frac{-13.6}{n^2} \text{ eV}$$

Calculated Energy Levels

Derived the expression for energy of electron in n th orbit: $E_n = -\frac{13.6}{n^2} \text{ eV}$ which matches experimental results.



Foundation for Quantum Theory

Laid the foundation for modern quantum theory and atomic models.

LIMITATIONS



Applicable Only to Hydrogen-like Atoms

Cannot explain spectra of multi-electron atoms.



Fails to Explain Fine Structure

Does not account for splitting of spectral lines in magnetic or electric fields (Zeeman and Stark effect).



Contradicts Wave Nature of Electron

Failed to incorporate the wave nature (de Broglie concept) of electron.



No Explanation for Relative Intensities

Could not explain the relative intensities of spectral lines.

Bohr's model was a great step towards understanding atoms, but it was later replaced by quantum mechanical model (Heisenberg, Schrödinger) which could explain the behavior of all atoms.



Niels Bohr
(1885–1962)

BOHR'S THEORY: HYDROGEN ATOM

CALCULATION OF ATOMIC RADIUS AND ENERGY & SPECTRA OF HYDROGEN ATOM

"An expert is a person who has made all the mistakes that can be made in a very narrow field."
– Niels Bohr

1. BOHR'S POSTULATES (FOR H-ATOM)

- Electrons revolve about the nucleus in certain stable circular orbits (stationary states) without radiating energy.

Allowed orbits
($n = 1, 2, 3, \dots$)

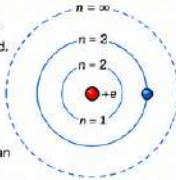
- Only those orbits are allowed for which the angular momentum of electron is quantized.

$$mvr = n \frac{h}{2\pi} \quad (n = 1, 2, 3, \dots)$$

- Radiation is emitted or absorbed only when an electron jumps from one stationary orbit to another. The frequency (ν) of radiation is given by:

$$h\nu = |E_i - E_f|$$

h = Planck's constant = 6.626×10^{-34} J s
 m = mass of electron = 9.109×10^{-31} kg
 e = charge on electron = 1.602×10^{-19} C
 ϵ_0 = permittivity of free space = 8.854×10^{-12} C² N⁻¹ m⁻²
 $k = \frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9$ Nm² C⁻²



2. CALCULATION OF RADIUS OF n^{th} ORBIT

For the electron in a circular orbit of radius r_n :

Centripetal force = Coulombic force

$$\frac{mv^2}{r_n} = k \frac{Ze^2}{r_n^2} \quad (Z = \text{atomic number})$$

From quantization of angular momentum,

$$mvr_n = n \frac{h}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr_n}$$

Substituting v in the first equation,

$$\frac{m}{r_n} \left(\frac{n^2 h^2}{4\pi^2 m^2 r_n^2} \right) = k \frac{Ze^2}{r_n^2}$$
$$r_n = \frac{n^2 h^2}{4\pi^2 m k Ze^2}$$

For Hydrogen ($Z = 1$):
 $r_n = a_0 n^2$

$$\text{where } a_0 = \frac{h^2}{4\pi^2 m k e^2} = 5.29 \times 10^{-11} \text{ m}$$

(Bohr radius)

Radii of some orbits in H-atom

n	1	2	3	4
r_n (m)	5.29×10^{-11}	2.12×10^{-10}	4.76×10^{-10}	8.46×10^{-10}

3. CALCULATION OF ENERGY OF n^{th} ORBIT

Total energy (E) of electron in n^{th} orbit = Kinetic energy (K) + Potential energy (U)

Kinetic energy,

$$K = \frac{1}{2} mv^2 = \frac{kZe^2}{2r_n}$$

Potential energy,

$$U = -\frac{kZe^2}{r_n}$$

$$\text{Subs. } r_n = \frac{n^2 h^2}{4\pi^2 m k Ze^2}$$
$$E_n = -\frac{2\pi^2 m k^2 Z^2 e^4}{h^2} \frac{1}{n^2}$$

$$E_n = K + U = \frac{kZe^2}{2r_n} - \frac{kZe^2}{r_n} = -\frac{kZe^2}{2r_n}$$

For Hydrogen ($Z = 1$):

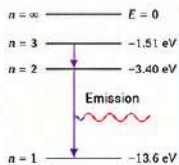
$$E_n = -\frac{13.6}{n^2} \text{ eV}$$

$$E_1 = -13.6 \text{ eV (ground state)}$$
$$E_2 = -3.40 \text{ eV}, E_3 = -1.51 \text{ eV}, E_4 = -0.85 \text{ eV}, \dots$$

Negative sign shows that electron is bound to the nucleus.
Energy becomes zero at $n = \infty$ (electron is free).

4. ENERGY TRANSITION

When electron jumps from higher orbit (n_i) to lower orbit (n_f), radiation is emitted.

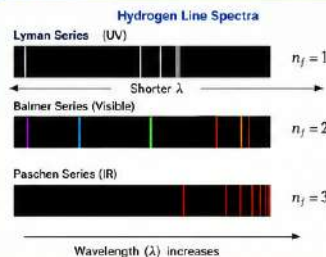


Energy of photon emitted
 $h\nu = E_i - E_f$
 $= 13.6 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{ eV}$
($n_i > n_f$)

5. SPECTRA OF HYDROGEN ATOM

Electron transitions ending at a particular lower level (n_f), produce a series of spectral lines.

Series	Final level (n_f)	Region	Color
Lyman series	1	Ultraviolet	(UV)
Balmer series	2	Visible	(VIBGYOR)
Paschen series	3	Infrared	(IR)
Brockett series	4	Infrared	
Pfund series	5	Infrared	



6. IMPORTANT FORMULAE (SUMMARY)

- Radius of n^{th} orbit (H-atom): $r_n = a_0 n^2$ where $a_0 = 5.29 \times 10^{-11}$ m
- Energy of n^{th} orbit (H-atom): $E_n = -\frac{13.6}{n^2} \text{ eV}$
- Energy of photon emitted/absorbed:
 $h\nu = 13.6 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{ eV}$
- Frequency: $\nu = \frac{c}{\lambda}$
- Rydberg formula for H-atom spectra:
 $\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$ where $R_H = 1.097 \times 10^7 \text{ m}^{-1}$

$n_i > n_f$
 R_H = Rydberg constant

In Short

Bohr's theory explains the stability of H-atom, quantized energy levels, calculation of atomic radius and energy, and the origin of line spectra.



Key Takeaway: Electrons occupy only certain allowed orbits. Energy is emitted/absorbed in jumps between orbits.

The strength of this theory lies in its simplicity, clarity and its ability to explain spectra.



Louis de Broglie
(1892–1987)

de BROGLIE DUAL NATURE OF MATTER

Matter exhibits both particle and wave nature

"If light can behave as matter, why not matter behave as light?"

1. HYPOTHESIS

Louis de Broglie (1924) proposed that **ALL MATTER** in motion has an associated **WAVE NATURE**.

Wavelength of matter wave (de Broglie wavelength)

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where,

h = Planck's constant (6.626×10^{-34} J s)

p = momentum of particle

m = mass

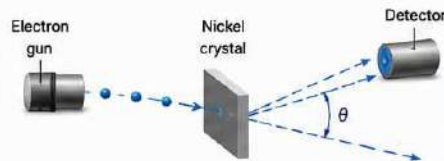
v = velocity

Key Points

- Greater momentum (p) → smaller wavelength (λ)
- For macroscopic objects (large m), λ is extremely small → wave nature is not observable.

2. EXPERIMENTAL EVIDENCE

Electron Diffraction – Davisson and Germer Experiment (1927)



Electrons show diffraction pattern just like X-rays, confirming their wave nature.

Calculated de Broglie wavelength

$$\lambda = \frac{h}{mv} = 1.67 \times 10^{-10} \text{ m}$$

This matched the wavelength obtained from the diffraction pattern.

Other Confirming Experiments

- G. P. Thomson (1927) – Diffraction of electrons by thin metal foil
- Reid & Planck (1927) – Diffraction and interference of electrons

3. DUAL NATURE – SUMMARY

Particle Nature

Matter behaves as particles in phenomena like:

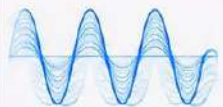
- Reflection
- Refraction
- Photoelectric effect
- Collision



Wave Nature

Matter behaves as waves in phenomena like:

- Diffraction
- Interference
- Polarization (in some cases)



Wave-Particle Duality

Matter (like light) shows both wave and particle properties depending on the experimental situation.

4. SIGNIFICANCE OF de BROGLIE DUAL NATURE OF MATTER



Unified Concept

Unified light (wave-particle duality) and matter under a common framework.



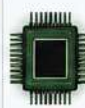
Foundation of Quantum Mechanics

Paved the way for the development of quantum theory and the wave equation (Schrödinger equation).



Explanation of Atomic Stability & Structure

Explains the standing wave nature of electrons in atoms, leading to quantized energy levels.



Basis of Modern Technology

Essential for the development of semiconductors, transistors, electron microscopes, etc.



Electron Microscopy

Wave nature of electrons is used in Electron Microscope for imaging ultra-small objects.



Quantum World Understanding

Helps us understand why microscopic particles behave differently from macroscopic objects.

In Short

de Broglie's hypothesis revolutionized physics by revealing that ALL MATTER has a wave nature.

This dual nature is a cornerstone of modern physics and has led to a wide range of scientific and technological advancements.

$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad (\text{de Broglie wavelength})$$



Werner Heisenberg
(1901–1976)

HEISENBERG UNCERTAINTY PRINCIPLE

You cannot know both the position and momentum of a particle exactly at the same time.

"Nature is not classical, dammit, and if you want to make a simulation of nature, you'd better make it quantum mechanical."

1. THE PRINCIPLE

It is impossible to simultaneously determine with perfect precision:

- Position (x) of a particle
- Momentum (p) of the particle

The more precisely we know one, the less precisely we can know the other.

Mathematical Expression

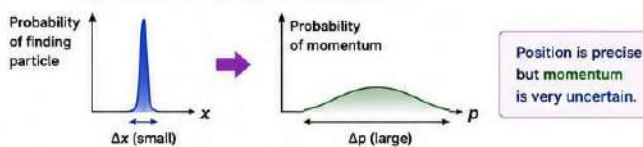
Heisenberg Uncertainty Principle

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

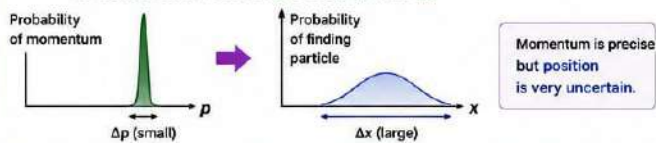
where, Δx = uncertainty in position
 Δp = uncertainty in momentum
 $\hbar = \frac{h}{2\pi}$ (reduced Planck's constant)
 $= 1.054 \times 10^{-34} \text{ J s}$

2. WHAT DOES IT MEAN?

If position is known precisely (small Δx)



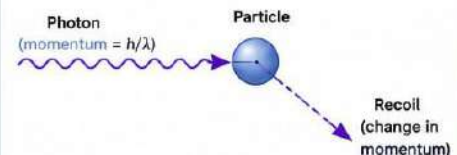
If momentum is known precisely (small Δp)



Nature does not allow both to be known with arbitrary precision.

3. WHY DOES IT HAPPEN?

Measuring position requires light (photon) to locate the particle.



Photon collides with the particle and alters its momentum (recoil). Thus, the act of measurement disturbs the particle — making exact knowledge impossible.

4. SIGNIFICANCE OF HEISENBERG UNCERTAINTY PRINCIPLE

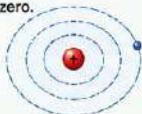
Foundation of Quantum Mechanics

Shows that quantum world is fundamentally different from the classical world.



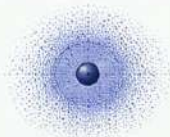
Explains Atomic Stability

Electrons in an atom cannot collapse into the nucleus because their position and momentum cannot both be zero.



Basis of Electron Cloud Concept

Electrons are described in terms of probability clouds (orbitals), not precise paths.



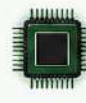
Sets Limits to Measurement

Defines a fundamental limit to the precision of any measurement in nature.



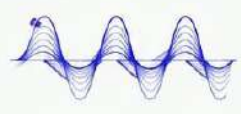
Important in Modern Technology

Essential in tools like electron microscopes, semiconductors, and nanotechnology.



Quantum World Understanding

Helps us understand why particles behave in strange ways at microscopic scales.



In Short

Heisenberg Uncertainty Principle tells us that nature has an inbuilt limit on how precisely we can know a particle's position and momentum at the same time. It is a cornerstone of quantum theory and explains the probabilistic nature of the microscopic world.

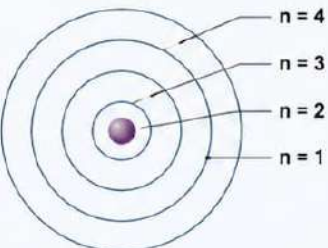
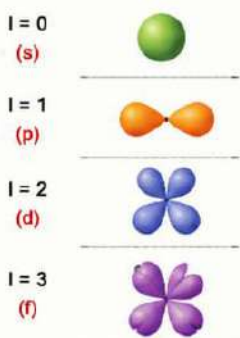
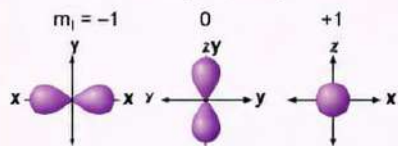
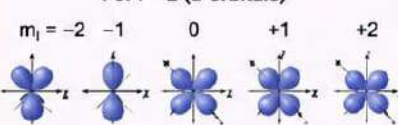
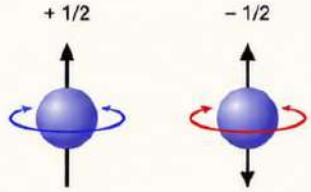
Remember the Principle:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

The more you know about where the particle is, the less you know about how fast it is moving!

QUANTUM NUMBERS AND THEIR SIGNIFICANCE

Quantum numbers describe the state of an electron in an atom. There are **four quantum numbers**.

1. PRINCIPAL QUANTUM NUMBER (n)	2. AZIMUTHAL (ORBITAL) QUANTUM NUMBER (l)	3. MAGNETIC QUANTUM NUMBER (m _l)	4. SPIN QUANTUM NUMBER (m _s)
Defines the main energy level or shell of the electron. $n = 1, 2, 3, 4, \dots \infty$  - Determines the size of the orbital and energy of the electron. - As n increases, energy and distance from nucleus increase.	Defines the subshell or shape of the orbital. $l = 0 \text{ to } (n - 1)$ $l = 0$ (s) $l = 1$ (p) $l = 2$ (d) $l = 3$ (f)  - Determines the shape of the orbital. - Number of subshells in a shell = n	Defines the orientation of the orbital in space. $m_l = -l \text{ to } 0 \text{ to } +l$ For $l = 1$ (p-orbitals) $m_l = -1 \quad 0 \quad +1$  For $l = 2$ (d-orbitals) $m_l = -2 \quad -1 \quad 0 \quad +1 \quad +2$  - Determines the spatial orientation of the orbital. - Number of orbitals in a subshell = $2l + 1$	Defines the spin of the electron in an orbital. $m_s = +\frac{1}{2} \text{ or } -\frac{1}{2}$ $+1/2$ $-1/2$  - Electrons in the same orbital must have opposite spins. - Explains electron pairing in orbitals.

SIGNIFICANCE OF QUANTUM NUMBERS

- They uniquely identify an orbital and the state of an electron in an atom.
- They help in writing the electronic configuration of elements.
- They explain the shape, size, energy, orientation and spin of orbitals.
- They are fundamental to understanding chemical bonding and properties of elements.

EXAMPLE

For the electron in $3p_x$ orbital with spin up

n	l	m _l	m _s
= 3	= 1 (p subshell)	= -1 (p _x orbital)	= $+\frac{1}{2}$ (spin up)

Set of four quantum numbers = $(3, 1, -1, +\frac{1}{2})$

Each electron in an atom is described by a unique set of four quantum numbers.

QUICK SUMMARY

Quantum Number	Symbol	Possible Values	What it Describes
Principal	n	1, 2, 3, ... ∞	Energy level, size of orbital
Azimuthal	l	0 to n - 1	Shape of orbital (s, p, d, f)
Magnetic	m _l	-l to 0 to +l	Orientation of orbital in space
Spin	m _s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Spin of electron (up or down)



Erwin Schrödinger
(1887–1961)

SHAPES OF ATOMIC ORBITALS

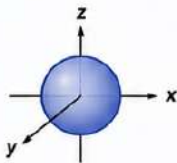
Atomic orbitals are regions around the nucleus where the probability of finding an electron is maximum.

"The wave function provides the most complete description of the state of a system."
– E. Schrödinger

1. s – ORBITALS

s orbitals are spherical in shape.

Number of s orbitals in each shell = 1



No angular nodes.

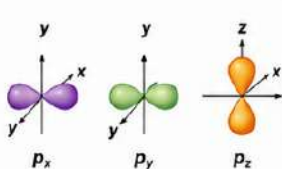
Orbital	1s	2s	3s
Shape			
Relative Size	Smallest	Larger	Larger

• As n increases, size of s orbital increases.

2. p – ORBITALS

p orbitals are dumbbell shaped.

Number of p orbitals in each shell = 3 (p_x, p_y, p_z)



One angular (nodal) plane.
(Plane of node passes through nucleus)

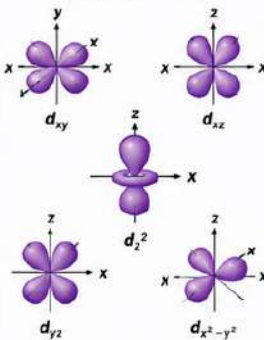
Orientation in space

p_x : Along x-axis
 p_y : Along y-axis
 p_z : Along z-axis

3. d – ORBITALS

d orbitals have more complex shapes.

Number of d orbitals in each shell = 5

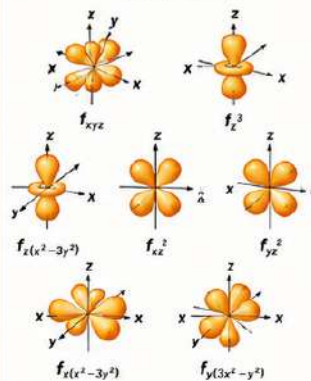


Two angular (nodal) planes for $d_{xy}, d_{xz}, d_{yz}, d_{x^2-y^2}$
One conical (nodal) surface for d_{z^2}

4. f – ORBITALS

f orbitals have even more complex shapes.

Number of f orbitals in each shell = 7



Three or more angular (nodal) planes or complex nodal surfaces.

5. NUMBER OF NODES

Nodes are regions where probability of finding an electron is zero.

Type of Orbital	Angular Nodes	Radial Nodes
s	0	$n - 1$
p	1	$n - 2$
d	2	$n - 3$
f	3	$n - 4$

Total number of nodes
= Angular nodes + Radial nodes
= $(l) + (n - l - 1) = n - 1$

where n = principal quantum number
l = azimuthal quantum number

6. SIGNIFICANCE

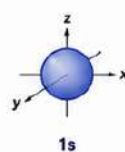
- Shapes of atomic orbitals help in understanding the electronic structure of atoms.
- They determine the direction of chemical bonding.
- They explain the shapes of molecules (VSEPR theory).
- They help in predicting magnetic properties of substances.
- Essential for understanding the structure of complex ions and molecular orbitals.

7. EXAMPLES OF ORBITALS IN ATOMS

Hydrogen (Z = 1)

Electronic configuration: $1s^1$

Only 1s orbital present.

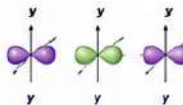


1s

Carbon (Z = 6)

Electronic configuration: $1s^2 2s^2 2p^2$

2p orbitals (p_x, p_y, p_z)



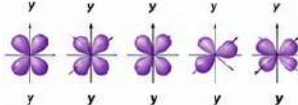
Electron occupancy

(Hund's rule followed)

Iron (Z = 26)

Electronic configuration: $[Ar] 3d^6 4s^2$

3d orbitals



(Maximum multiplicity)

8. QUICK SUMMARY TABLE

Orbital Type	Shape	No. of Orbitals per Subshell	Orientation
s	Spherical	1	No specific orientation
p	Dumbbell	3	Along x, y, z axes
d	Complex (4-lobed, etc.)	5	Between axes and along axes
f	Complex (multi-lobed)	7	Various orientations

In Short

- s orbitals are spherical, p orbitals are dumbbell shaped, d and f orbitals have complex shapes.
- Number of orbitals in a subshell = $2l + 1$



Remember:

Number of orbitals in a subshell = $2l + 1$
s ($l = 0$) → 1 orbital p ($l = 1$) → 3 orbitals
d ($l = 2$) → 5 orbitals f ($l = 3$) → 7 orbitals

Key Takeaway:

Shape and orientation of atomic orbitals are determined by the azimuthal quantum number (l) and are fundamental to the structure and reactivity of atoms and molecules.



Niels Bohr
(1885–1962)

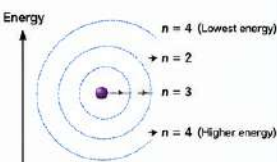
ENERGIES OF ATOMIC ORBITALS

The energy of an orbital depends on the principal quantum number (n) and (for same n) on the azimuthal quantum number (l).

"The energies of atomic orbitals determine the electronic structure, stability and reactivity of atoms." – N. Bohr

1. BASIC IDEA

Electrons in an atom occupy orbitals of definite energies. The energy of an orbital increases with increase in distance from the nucleus. Thus, orbitals farther from the nucleus have higher energy.



Key Point
Energy $\propto$ Distance from nucleus

2. ORDER OF ORBITAL ENERGIES (SAME PRINCIPAL SHELL, n)

For orbitals having the same n but different l , energy increases with increase in l .

Order of energy: $s < p < d < f$

Subshell (l)	Value of l	Energy	Shape
s	0	Lowest	
p	1	Higher	
d	2	Higher	
f	3	Highest	

Reason: As l increases, electron spends more time farther from the nucleus $\rightarrow$ higher energy.

3. ORDER OF ORBITAL ENERGIES (DIFFERENT PRINCIPAL SHELLS)

For orbitals having different n values, energy depends on both n and l .

General trend:

Lower $n \rightarrow$ Lower energy

For same n :

$s < p < d < f$

Madelung or ($n + l$) Rule:

For any two orbitals, the one with lower ($n + l$) has lower energy.

If ($n + l$) is same, the orbital with lower n has lower energy.

Examples:

4s ($n + l = 4 + 0 = 4$) < 3d ($n + l = 3 + 2 = 5$)

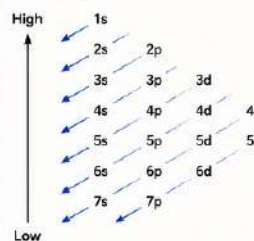
So, 4s is lower in energy than 3d.

4p ($4 + 1 = 5$) < 3d ($3 + 2 = 5$)

Same ($n + l$), but $3 < 4 \rightarrow$ 3d is lower.

4. ENERGY DIAGRAM (AUFBAU ORDER)

Relative energies of orbitals (following Aufbau principle):



(Diagonal rule gives the order of increasing energy of orbitals.)

5. RELATIVE ENERGY OF SUBSHELLS

Energy increases in the order:

$n = 1 : 1s$

$n = 2 : 2s < 2p$

$n = 3 : 3s < 3p < 3d$

$n = 4 : 4s < 4p < 4d < 4f$

$n = 5 : 5s < 5p < 5d < 5f$

$n = 6 : 6s < 6p < 6d$

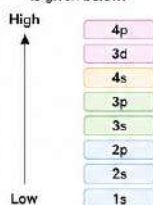
$n = 7 : 7s < 7p$

Note

In multi-electron atoms, energies of orbitals are affected by electron-electron repulsion and shielding, though the above order remains generally valid.

6. ENERGY COMPARISON OF SOME ORBITALS

A comparison of energies of some orbitals is given below:



Important Comparisons:

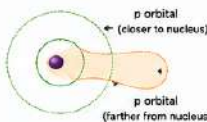
- $4s < 3d < 4p$
- $5s < 4d < 5p < 5d$
- $6s < 4f < 5d < 6p$ (approximate order)

7. WHY DO ORBITALS HAVE DIFFERENT ENERGIES?

✓ Electrons in different orbitals experience different effective nuclear charge (Z_{eff}).

✓ Orbitals with higher l penetrate less into the nucleus and experience lower attraction.

✓ Thus, they have higher energy.



Greater the average distance of an electron from nucleus, higher is the energy of the orbital.

8. SIGNIFICANCE



Determines the order of filling electrons in orbitals (Aufbau principle).



Explains the stability and structure of atoms.



Helps in understanding magnetic properties.



Useful in predicting reactivity and chemical behaviour of elements.

QUICK SUMMARY

Factor	Effect on Energy	Order
Principal quantum number (n)	As n increases, energy increases	$1 < 2 < 3 < \dots$
Azimuthal quantum number (l) (same n)	As l increases, energy increases	$s < p < d < f$
($n + l$) Rule	Lower ($n + l$) $\rightarrow$ lower energy	—
If ($n + l$) same	Lower $n \rightarrow$ lower energy	—

In Short

Energy of an orbital increases with increase in n and (for same n) with increase in l . The ($n + l$) rule helps in comparing the energies of orbitals in multi-electron atoms.



Remember:

- Lower $n \rightarrow$ Lower energy
- For same n : $s < p < d < f$
- Lower ($n + l$) $\rightarrow$ Lower energy
- If ($n + l$) same, lower $n \rightarrow$ Lower energy

Key Takeaway:

The arrangement of orbitals in order of increasing energy is the foundation of electronic configuration and periodic properties of elements.

AUFBAU PRINCIPLE

Orderly filling of atomic orbitals in the ground state of an atom

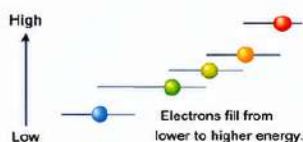
1. STATEMENT

Electrons occupy the available orbitals in order of increasing energy.

In the ground state of an atom, electrons fill the lowest energy orbitals first and then proceed to higher energy orbitals.

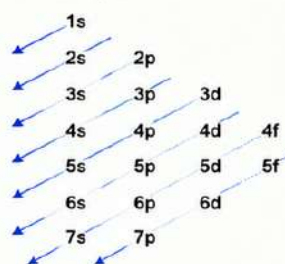
Key Point

- Electrons always enter the lowest energy orbital available.



2. ENERGY ORDER OF ORBITALS (AUFBAU DIAGONAL RULE)

The relative energies of orbitals increase in the following order:



Fill orbitals in the order of the arrows (diagonally from top right to bottom left).

3. RULES FOLLOWED

- Fill the lowest energy orbital first.
- Each orbital can accommodate maximum two electrons with opposite spins.
- Hund's rule is followed in orbitals of equal energy (like p, d, f orbitals).
- Pauli's exclusion principle is obeyed.

These rules ensure:

- Minimum energy (maximum stability)
- Proper distribution of electrons
- Unique electronic configuration

4. ORBITAL CAPACITY

Maximum number of electrons in a subshell:

- s → 1 orbital → 2 electrons
- p → 3 orbitals → 6 electrons
- d → 5 orbitals → 10 electrons
- f → 7 orbitals → 14 electrons

Maximum electrons in a shell ($2n^2$)

Shell (n)	Max. electrons
1	2
2	8
3	18
4	32
5	50
6	72
7	98

5. EXAMPLES OF ELECTRONIC CONFIGURATION BY AUFBAU PRINCIPLE

Example 1: Hydrogen (Z = 1)

1 electron

Configuration: $1s^1$

Orbital diagram:

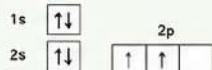


Example 2: Carbon (Z = 6)

6 electrons

Configuration: $1s^2 2s^2 2p^2$

Orbital diagram:



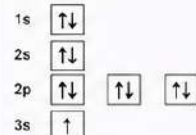
Hund's rule: electrons occupy degenerate orbitals singly with parallel spins.

Example 3: Sodium (Z = 11)

11 electrons

Configuration: $1s^2 2s^2 2p^6 3s^1$

Orbital diagram:



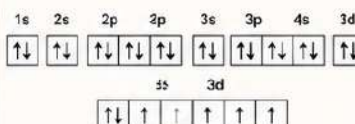
Example 4: Iron (Z = 26)

26 electrons

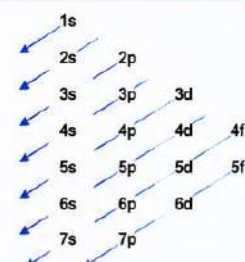
Configuration:

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

Orbital diagram:



6. AUFBAU DIAGRAM (SUMMARY)



7. SIGNIFICANCE

- Provides a simple and logical method to write electronic configuration.
- Explains the stability of atoms in ground state.
- Basis for understanding periodic properties and chemical behaviour.
- Helps in predicting valency, oxidation states and magnetic properties.

8. LIMITATIONS

- Does not apply to some elements like Cr (24), Cu (29), Ag (47), Au (79).
- These exceptions arise due to extra stability of half-filled or completely filled subshells.
- Actual energy order of orbitals in multi-electron atoms may change slightly.

9. EXCEPTIONS (DUE TO EXTRA STABILITY)

Some configurations are more stable than predicted by Aufbau principle.

Element (Z)	Aufbau Prediction	Actual Configuration	Reason
Cr (24)	$[\text{Ar}] 4s^2 3d^4$	$[\text{Ar}] 4s^1 3d^5$	Half-filled d subshell (extra stable)
Cu (29)	$[\text{Ar}] 4s^2 3d^9$	$[\text{Ar}] 4s^1 3d^{10}$	Completely filled d subshell
Ag (47)	$[\text{Kr}] 5s^2 4d^9$	$[\text{Kr}] 5s^1 4d^{10}$	Completely filled d subshell
Au (79)	$[\text{Xe}] 6s^2 4f^{14} 5d^9$	$[\text{Xe}] 6s^1 4f^{14} 5d^{10}$	Completely filled d subshell

In Short

Aufbau principle states that electrons fill the orbitals in order of increasing energy. It is the foundation for the electronic configuration of elements.



Remember: Fill from lower to higher energy following the Aufbau order.

Key Order:

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s...$



Wolfgang Pauli
(1900–1958)

PAULI EXCLUSION PRINCIPLE

No two electrons in an atom can have the same set of four quantum numbers.

"This principle is the foundation of the structure of atoms and the entire periodic table."
– Wolfgang Pauli

1. STATEMENT

No two electrons in an atom can have the same set of **four quantum numbers** (n, l, m_l, m_s).

In simple words:

An orbital can accommodate maximum two electrons and they must have **opposite spins**.

- ✓ This principle was proposed by Wolfgang Pauli in 1925.
- ✓ It arises because electrons are fermions and obey the spin statistics.

2. ORBITAL OCCUPANCY

Each orbital can hold a maximum of two electrons with opposite spins.

Representation:

Allowed	Not allowed
$\uparrow\downarrow$ (Opposite spins)	$\uparrow\uparrow$ or $\downarrow\downarrow$ (Same spins)

Examples:

1s orbital: $[\uparrow\downarrow]$ (Allowed)
Not allowed: $[\uparrow\uparrow]$ or $[\downarrow\downarrow]$

2p_x orbital: $[\uparrow\downarrow]$ (Allowed)
Not allowed: $[\downarrow\downarrow]$ or $[\uparrow\uparrow]$

3. WHY IS IT NECESSARY?

- Electrons are indistinguishable particles.
- If two electrons had the same four quantum numbers, they would be identical.
- This would violate the basic rules of quantum mechanics.
- Due to Pauli Exclusion Principle:
 - Electrons occupy different orbitals
 - Atoms are stable
 - Periodic table exists



It is a fundamental principle that governs the electronic configuration and chemical behaviour of elements.

4. EFFECT ON ELECTRONIC CONFIGURATION

- Electrons fill available orbitals in order of increasing energy (Aufbau principle).
- No two electrons in the same orbital can have the same spin.
- Electrons pair up in an orbital only after singly occupying it with parallel spins (Hund's rule).

Example: Carbon (Z = 6)

Electronic configuration: $1s^2 2s^2 2p^2$

Correct (Pauli followed)

1s	2s	2p _x	2p _y	2p _z
$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$	$\uparrow$	

Incorrect (Pauli violated)

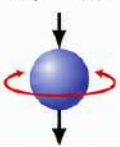
1s	2s	2p _x	2p _y	2p _z
$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow\downarrow$		

5. VISUALIZING SPIN IN AN ORBITAL

Spin up
($m_s = +1/2$)



Spin down
($m_s = -1/2$)



Thus, two electrons in the same orbital must have opposite spins.

6. EXAMPLES OF PAULI EXCLUSION PRINCIPLE

Example 1: Helium (Z = 2)

Electronic configuration: $1s^2$

Orbital diagram:

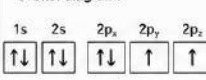


Two electrons in 1s orbital with opposite spins.

Example 2: Oxygen (Z = 8)

Electronic configuration: $1s^2 2s^2 2p^4$

Orbital diagram:

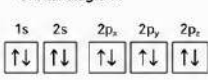


Each orbital has at most two electrons with opposite spins.

Example 3: Neon (Z = 10)

Electronic configuration: $1s^2 2s^2 2p^6$

Orbital diagram:



All orbitals are completely filled with opposite spins.

7. SIGNIFICANCE



Explains the stability of atoms and uniqueness of electronic configuration.



Leads to the existence of the periodic table.



Foundation for chemical bonding, magnetism and modern electronic devices.



Essential for understanding the behaviour of matter at atomic and molecular level.

In Short

Pauli Exclusion Principle states that no two electrons in an atom can have the same set of four quantum numbers (n, l, m_l, m_s).
An orbital can hold maximum two electrons with opposite spins.



Remember: No same four quantum numbers
→ No identical electrons
→ Opposite spins in same orbital

Key Takeaway:

This principle is at the heart of quantum mechanics and the structure of atoms.



Friedrich Hund
(1896–1997)

HUND'S RULE OF MAXIMUM MULTIPLICITY

Electrons occupy degenerate orbitals singly with parallel spins to maximize multiplicity.

"The ground state of an atom has the maximum possible multiplicity."
– F. Hund

1. STATEMENT

In a set of degenerate orbitals (orbitals having the same energy), electrons enter each orbital singly with parallel spins before pairing starts.

In simple words:

Electrons first occupy all degenerate orbitals one by one with the same (spin up) direction.

Only after that, pairing of electrons (with opposite spin) begins.

- ✓ Proposed by Friedrich Hund in 1925.
- ✓ Also called Rule of Maximum Multiplicity.

2. WHY IS IT NECESSARY?

- Minimizes electron–electron repulsion.
- Maximizes total spin (multiplicity) of the system.
- Leads to greater stability of the atom in the ground state.

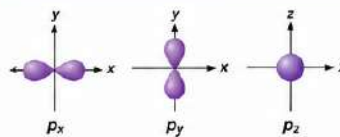


It is a fundamental principle that explains the observed electronic configurations of atoms.

3. HOW DOES IT WORK?

- 1 Electrons enter degenerate orbitals (same energy) one by one.
- 2 Each electron has the same (spin up) direction.
- 3 Pairing (opposite spins) starts only after all orbitals are singly occupied.

Example: p orbitals (p_x , p_y , p_z) are degenerate (same energy).



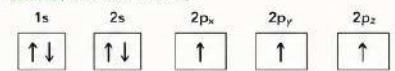
4. EFFECT ON ELECTRONIC CONFIGURATION

- Ensures maximum stability of atoms.
- Determines actual electronic configuration of many elements.
- Affects magnetic properties of atoms and ions.

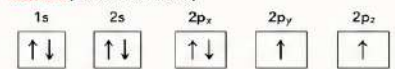
Example: Nitrogen ($Z = 7$)

Electronic configuration: $1s^2 2s^2 2p^3$

Correct (Hund's rule followed)



Incorrect (Hund's rule violated)

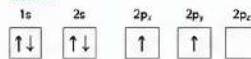


5. EXAMPLES OF HUND'S RULE IN ACTION

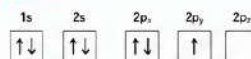
Example 1: Carbon ($Z = 6$)

Electronic configuration: $1s^2 2s^2 2p^2$

Correct:



Incorrect:

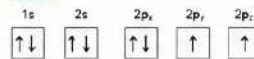


Electrons occupy $2p_x$ and $2p_y$ singly before pairing.

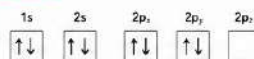
Example 2: Oxygen ($Z = 8$)

Electronic configuration: $1s^2 2s^2 2p^4$

Correct:



Incorrect:



After singly occupying all three p orbitals, fourth electron pairs up.

Example 3: Phosphorus ($Z = 15$)

Electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^3$

Correct:



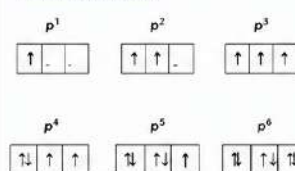
Incorrect:



Three electrons occupy $3p$ orbitals singly with parallel spins.

6. VISUAL SUMMARY (p SUBSHELL)

As electrons are added:



Electrons fill degenerate orbitals singly with parallel spins before pairing.

7. SIGNIFICANCE



Explains stability of atomic ground state.



Helps in predicting electronic configuration.



Basis for understanding magnetic behaviour (paramagnetism).



Essential for explaining periodicity and chemical properties of elements.

In Short

Hund's Rule states that in a set of degenerate orbitals, electrons enter singly with parallel spins to maximize multiplicity and stability. Pairing occurs only after all orbitals are singly occupied.



Remember:

Singly first, parallel spins always;
Pairing only when required.

Key Takeaway:

Maximize multiplicity → Maximize stability.



Gilbert N. Lewis
(1875–1946)
Pioneer in Chemical Bonding
and Electronic Structure

ELECTRONIC CONFIGURATION OF ATOMS

Distribution of electrons in various orbitals of an atom in the ground state.

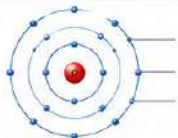
"Atoms build up by adding electrons one at a time. Their properties depend on the way these electrons are arranged."
— G. N. Lewis

1. WHAT IS ELECTRONIC CONFIGURATION?

It is the arrangement of electrons in different orbitals of an atom in its ground state.

Key Point

It shows how electrons are distributed in shells, subshells and orbitals.



Electrons occupy orbitals in order of increasing energy (Aufbau principle).

2. RULES FOR ELECTRONIC CONFIGURATION

- ✓ **Aufbau Principle ($n + l$ rule):** Electrons fill orbitals in order of increasing energy.
- ✓ **Pauli Exclusion Principle:** No two electrons in an orbital can have the same set of four quantum numbers.
- ✓ **Hund's Rule of Maximum Multiplicity:** Electrons occupy degenerate orbitals singly with parallel spins before pairing.
- ✓ **Maximum Capacity of Orbitals:**
 $s \rightarrow 2 e^-$ $p \rightarrow 6 e^-$ $d \rightarrow 10 e^-$ $f \rightarrow 14 e^-$

Order of filling (Aufbau order):

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d < 7p$

3. METHOD OF WRITING CONFIGURATION

- 1 Find atomic number (Z) $\rightarrow$ total electrons.
- 2 Write orbitals in Aufbau order.
- 3 Fill electrons as per rules:
 - Aufbau principle
 - Pauli exclusion principle
 - Hund's rule
- 4 Write the configuration using superscripts (e.g., $1s^2 2s^2 2p^6 \dots$).



Electronic configuration represents the ground state of an atom.

4. AUFBAU DIAGONAL RULE (TRICK)

The order of filling orbitals can be remembered using the diagonal rule.



Follow the arrows from top right to bottom left to get the order of increasing energy.

5. EXAMPLES OF ELECTRONIC CONFIGURATION

Example 1: Hydrogen ($Z = 1$)

Electronic configuration:

$1s^1$

Orbital diagram:



Only one electron in 1s orbital.

Example 2: Carbon ($Z = 6$)

Electronic configuration:

$1s^2 2s^2 2p^2$

Orbital diagram:



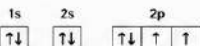
Hund's rule: Electrons occupy 2p orbitals singly with parallel spins.

Example 3: Oxygen ($Z = 8$)

Electronic configuration:

$1s^2 2s^2 2p^4$

Orbital diagram:



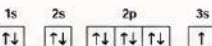
After singly occupying all three 2p orbitals, fourth electron pairs up.

Example 4: Sodium ($Z = 11$)

Electronic configuration:

$1s^2 2s^2 2p^6 3s^1$

Orbital diagram:



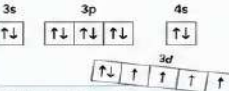
After 2p is filled, next electron enters 3s orbital.

Example 5: Iron ($Z = 26$)

Electronic configuration:

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

Orbital diagram:



3d has 6 electrons: five orbitals filled as $\uparrow\uparrow\uparrow\uparrow\uparrow$ then one pairs.

Number of electrons in shells

Shell (n)	Maximum electrons ($2n^2$)
1	2
2	8
3	18
4	32
5	50
6	72
7	98

6. SIGNIFICANCE

- ✓ Helps in understanding the structure and stability of atoms.
- ✓ Explains periodic properties of elements.
- ✓ Helps in predicting valency, oxidation states and magnetic behaviour.
- ✓ Basis for understanding chemical bonding and formation of ions and molecules.



7. EXCEPTIONS

Some elements do not follow the regular Aufbau order due to extra stability:

- ✗ **Chromium ($Z = 24$):**
Expected $\rightarrow [Ar] 4s^2 3d^4$
Actual $\rightarrow [Ar] 4s^1 3d^5$
- ✗ **Copper ($Z = 29$):**
Expected $\rightarrow [Ar] 4s^2 3d^9$
Actual $\rightarrow [Ar] 4s^1 3d^{10}$

Half-filled and fully-filled subshells are extra stable.

8. QUICK SUMMARY TABLE

Subshell	Number of Orbitals	Maximum Electrons	Shape
s	1	2	Spherical
p	3	6	Dumbbell
d	5	10	Complex (4-lobed)
f	7	14	Complex (multi-lobed)

9. KEY POINTS TO REMEMBER

- ✓ Electrons fill orbitals from lower to higher energy.
- ✓ No two electrons in the same orbital can have same four quantum numbers.
- ✓ Electrons occupy degenerate orbitals singly with parallel spins.
- ✓ Electronic configuration is unique for each atom.
- ✓ Valence electrons determine chemical behaviour.

In Short

Electronic configuration tells us how electrons are arranged in an atom. It follows specific rules (Aufbau, Pauli, Hund's) and helps in explaining the stability and properties of elements.



Remember:

Aufbau order + Pauli exclusion principle
+ Hund's rule = Correct electronic configuration

Key Takeaway:

Correct electronic configuration is the foundation of atomic structure, periodicity and chemical reactivity.